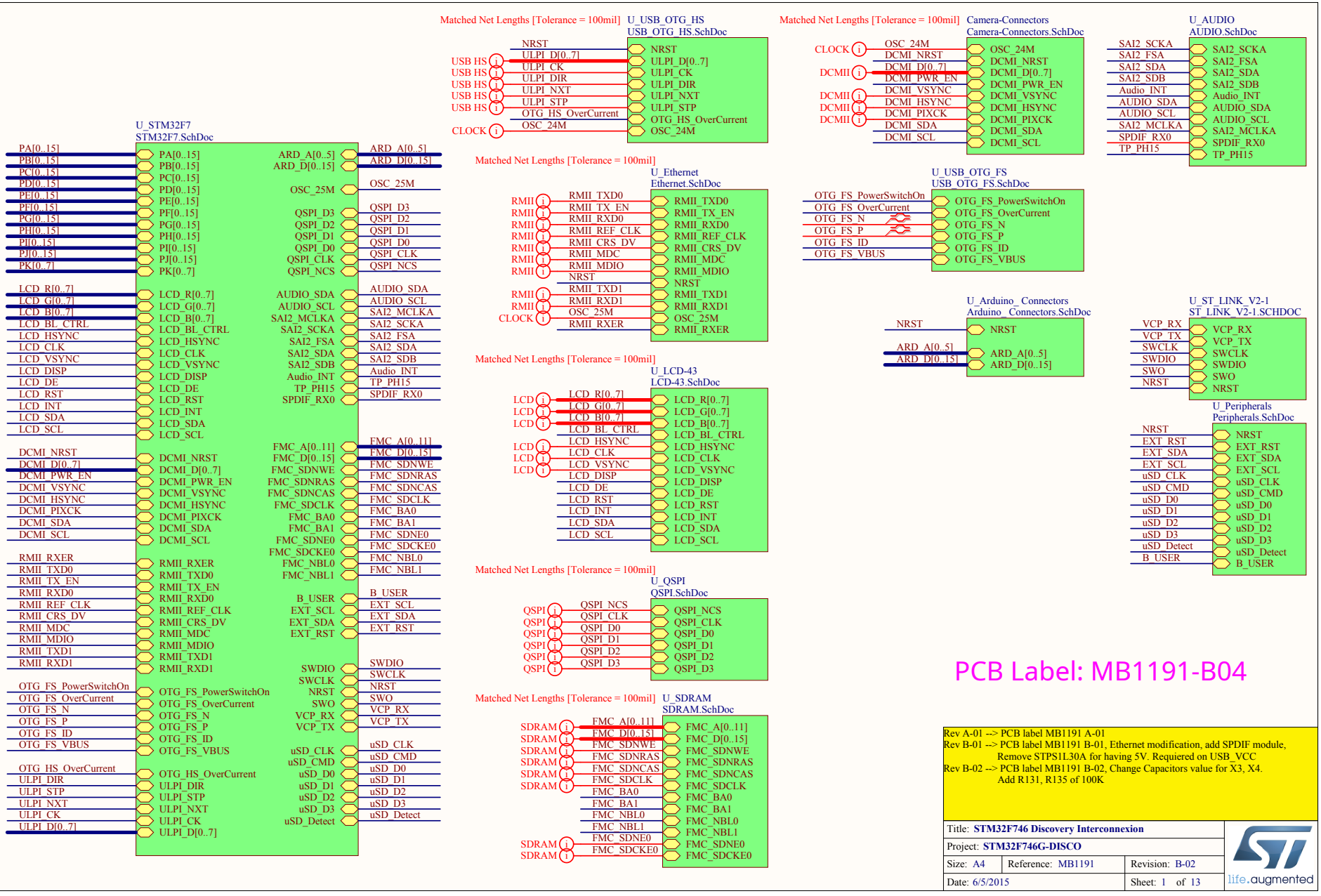
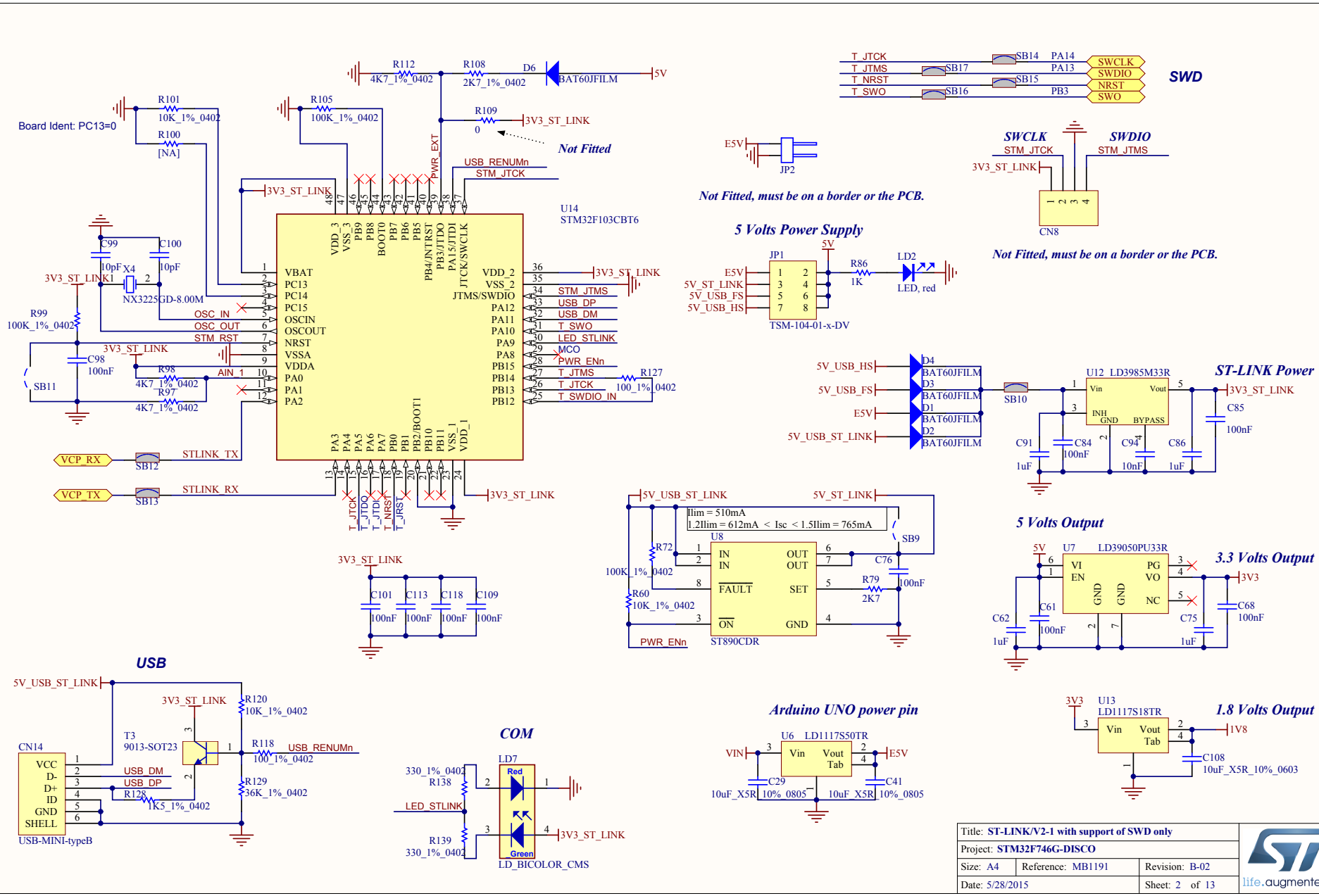
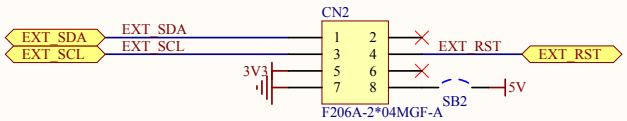

U_STM32F7
STM32F7.SchDoc
PA[0..15]
PB[0..15]
PC[0..15]
PD[0..15]
PE[0..15]
PF[0..15]
PG[0..15]
PH[0..15]
PI[0..15]
PJ[0..15]
PK[0..7]
LCD_R[0..7]
LCD_G[0..7]
LCD_B[0..7]
LCD_BL_CTRL
LCD_HSYNC
LCD_CLK
LCD_VSYNC
LCD_DISP
LCD_DE
LCD_RST
LCD_INT
LCD_SDA
LCD_SCL
DCMI_NRST
DCMI_D[0..7]
DCMI_PWR_EN
DCMI_VSYNC
DCMI_HSYNC
DCMI_PIXCK
DCMI_SDA
DCMI_SCL
RMII_RXER
RMII_TXD0
RMII_TX_EN
RMII_RXD0
RMII_REF_CLK
RMII_CRS_DV
RMII_MDC
RMII_MDIO
RMII_TXD1
RMII_RXD1
OTG_FS_PowerSwitchOn
OTG_FS_OverCurrent
OTG_FS_N
OTG_FS_P
OTG_FS_ID
OTG_FS_VBUS
OTG_HS_OverCurrent
ULPI_DIR
ULPI_STP
ULPI_NXT
ULPI_CK
ULPI_D[0..7]
ARD_A[0..5]
ARD_D[0..15]
OSC_25M
QSPI_D3
QSPI_D2
QSPI_D1
QSPI_D0
QSPI_CLK
QSPI_NCS
AUDIO_SDA
AUDIO_SCL
SAI2_MCLKA
SAI2_SCKA
SAI2_FSA
SAI2_SDA
SAI2_SDB
Audio_INT
TP_PH15
SPDIF_RX0
FMC_A[0..11]
FMC_D[0..15]
FMC_SDNWE
FMC_SDNRAS
FMC_SDNCAS
FMC_SDCLK
FMC_BA0
FMC_BA1
FMC_SDNE0
FMC_SDCKE0
FMC_NBL0
FMC_NBL1
B_USER
EXT_SCL
EXT_SDA
EXT_RST
SWDIO
SWCLK
NRST
SWO
VCP_RX
VCP_TX
uSD_CLK
uSD_CMD
uSD_D0
uSD_D1
uSD_D2
uSD_D3
uSD_Detect
Matched Net Lengths [Tolerance = 100mil]
U_USB_OTG_HS
USB_OTG_HS.SchDoc
USB HS
CLOCK
Matched Net Lengths [Tolerance = 100mil]
U_Ethernet
Ethernet.SchDoc
RMII
CLOCK
Matched Net Lengths [Tolerance = 100mil]
U_LCD-43
LCD-43.SchDoc
LCD
Matched Net Lengths [Tolerance = 100mil]
U_QSPI
QSPI.SchDoc
QSPI
Matched Net Lengths [Tolerance = 100mil]
U_SDRAM
SDRAM.SchDoc
SDRAM
Matched Net Lengths [Tolerance = 100mil]
Camera-Connectors
Camera-Connectors.SchDoc
CLOCK
DCMII
U_USB_OTG_FS
USB_OTG_FS.SchDoc
U_Arduino_ Connectors
Arduino_ Connectors.SchDoc
U_AUDIO
AUDIO.SchDoc
U_ST_LINK_V2-1
ST_LINK_V2-1.SCHDOC
U_Peripherals
Peripherals.SchDoc
PCB Label: MB1191-B04
Rev A-01 --> PCB label MB1191 A-01
Rev B-01 --> PCB label MB1191 B-01, Ethernet modification, add SPDIF module,
Remove STPS1L30A for having 5V. Requiered on USB_VCC
Rev B-02 --> PCB label MB1191 B-02, Change Capacitors value for X3, X4.
Add R131, R135 of 100K
Title: STM32F746 Discovery Interconnexion
Project: STM32F746G-DISCO
Size: A4
Reference: MB1191
Revision: B-02
Date: 6/5/2015
Sheet: 1 of 13
life.augmented

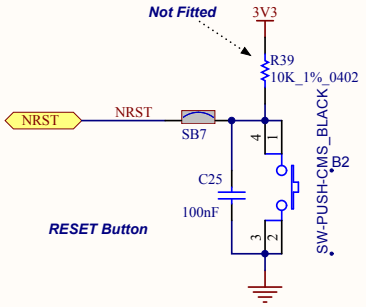


Title: **ST-LINK/V2-1 with support of SWD only**		
Project: **STM32F746G-DISCO**		
Size: A4	Reference: MB1191	Revision: B-02
Date: 5/28/2015		Sheet: 2 of 13

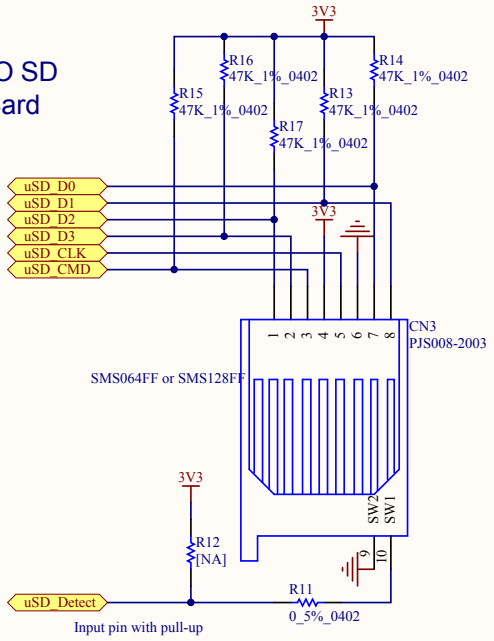
life.augmented

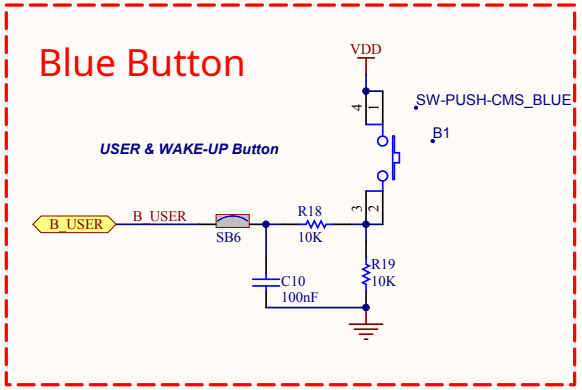
EXT/RF E2P Connector





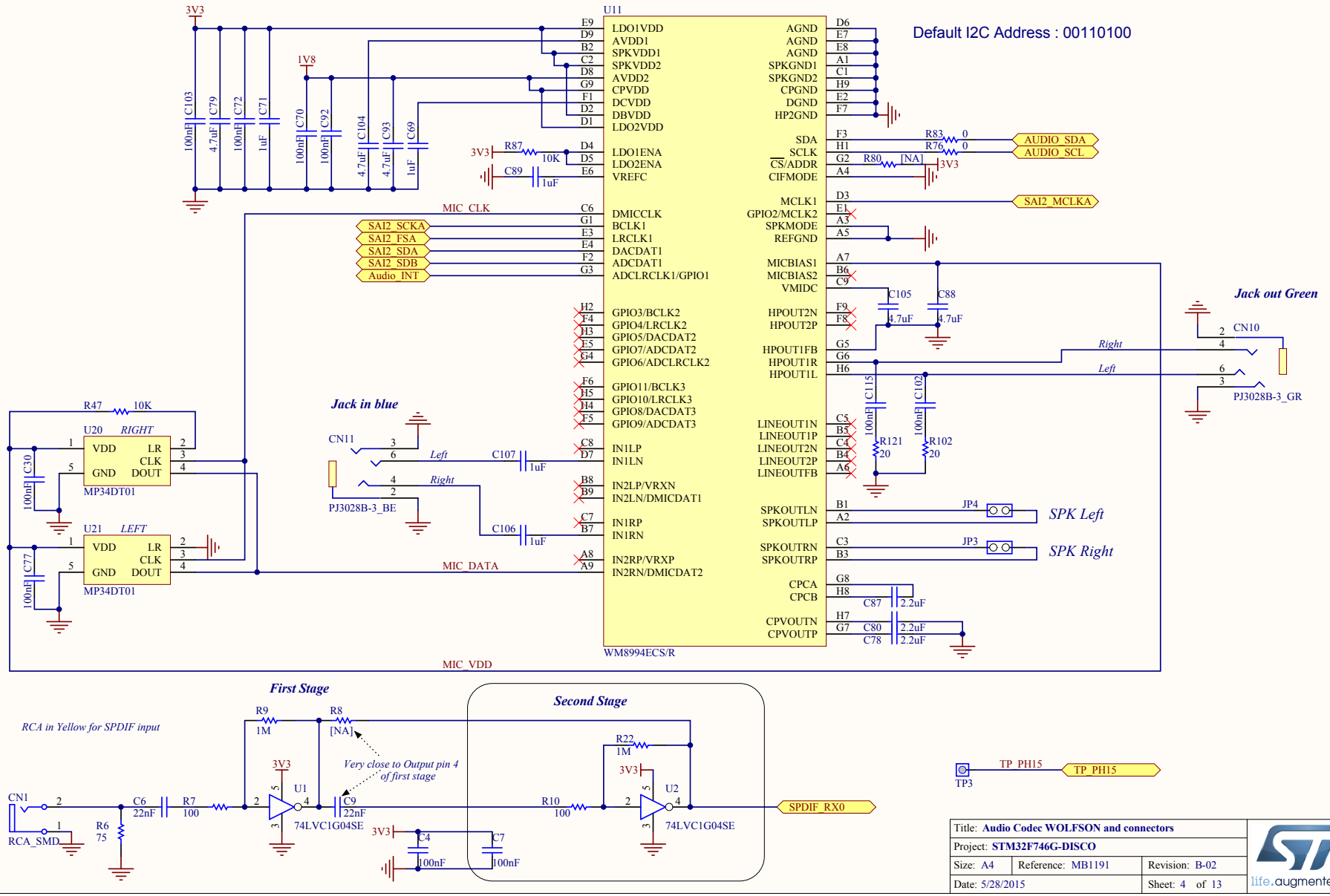
MICRO SD (TF) Card





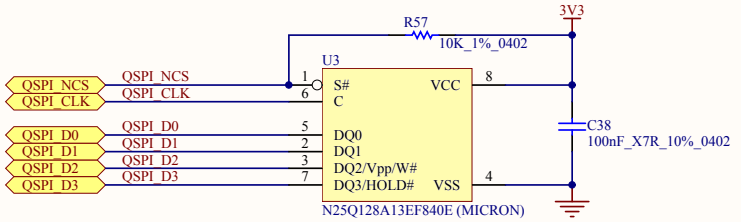
Title: **Joystick, ACP, LEDs and Push Button**			
Project: **STM32F746G-DISCO**			
Size: A4	Reference: MB1191	Revision: B-02	
Date: 5/28/2015		Sheet: 3 of 13	

life.augmented


Default I2C Address : 00110100
U11
WM8994ECS/R
3V3
1V8
MIC_CLK
MIC_DATA
MIC_VDD
SAI2_SCKA
SAI2_FSA
SAI2_SDA
SAI2_SDB
Audio_INT
AUDIO_SDA
AUDIO_SCL
SAI2_MCLKA
Jack out Green
CN10
PJ3028B-3_GR
Right
Left
Jack in blue
CN11
PJ3028B-3_BE
U20 RIGHT
U21 LEFT
MP34DT01
SPK Left
SPK Right
JP4
JP3
First Stage
Second Stage
RCA in Yellow for SPDIF input
Very close to Output pin 4 of first stage
CN1
RCA_SMD
74LVC1G04SE
SPDIF_RX0
TP_PH15
Title: Audio Codec WOLFSON and connectors
Project: STM32F746G-DISCO
Size: A4
Reference: MB1191
Revision: B-02
Date: 5/28/2015
Sheet: 4 of 13
life.augmented

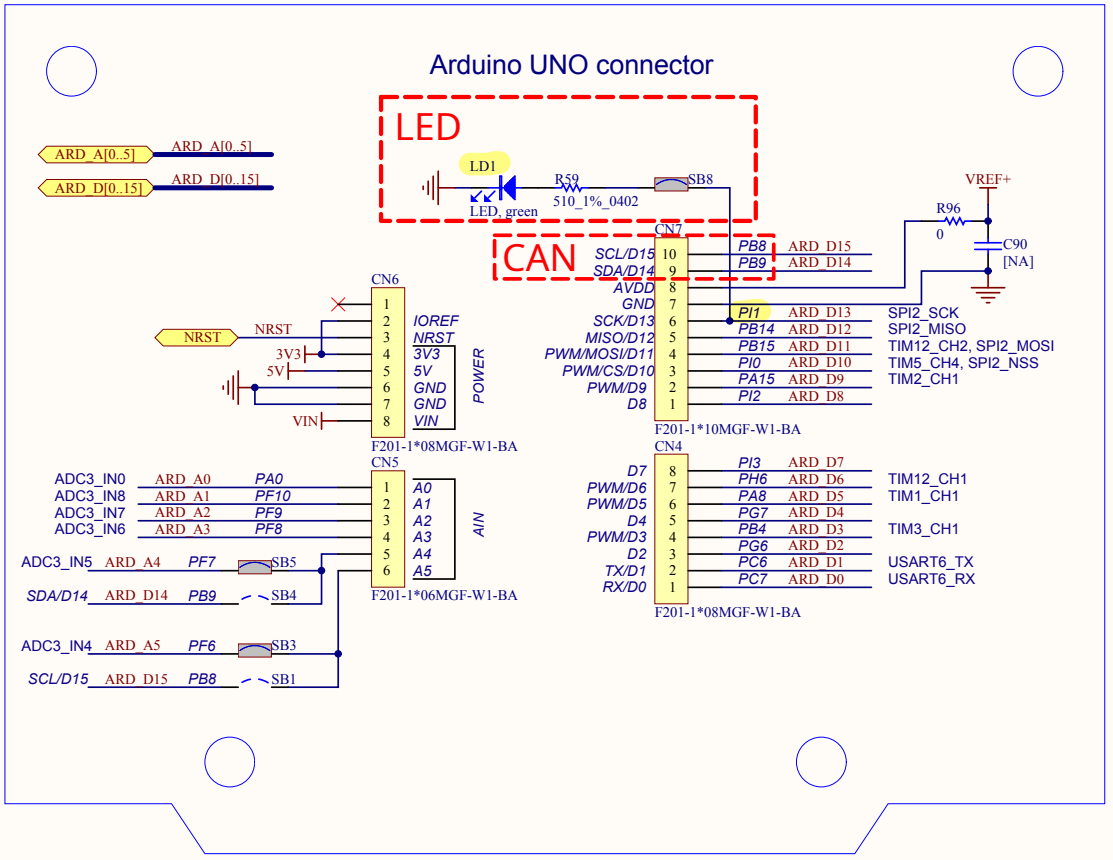
Memory

Quad SPI Flash Memory



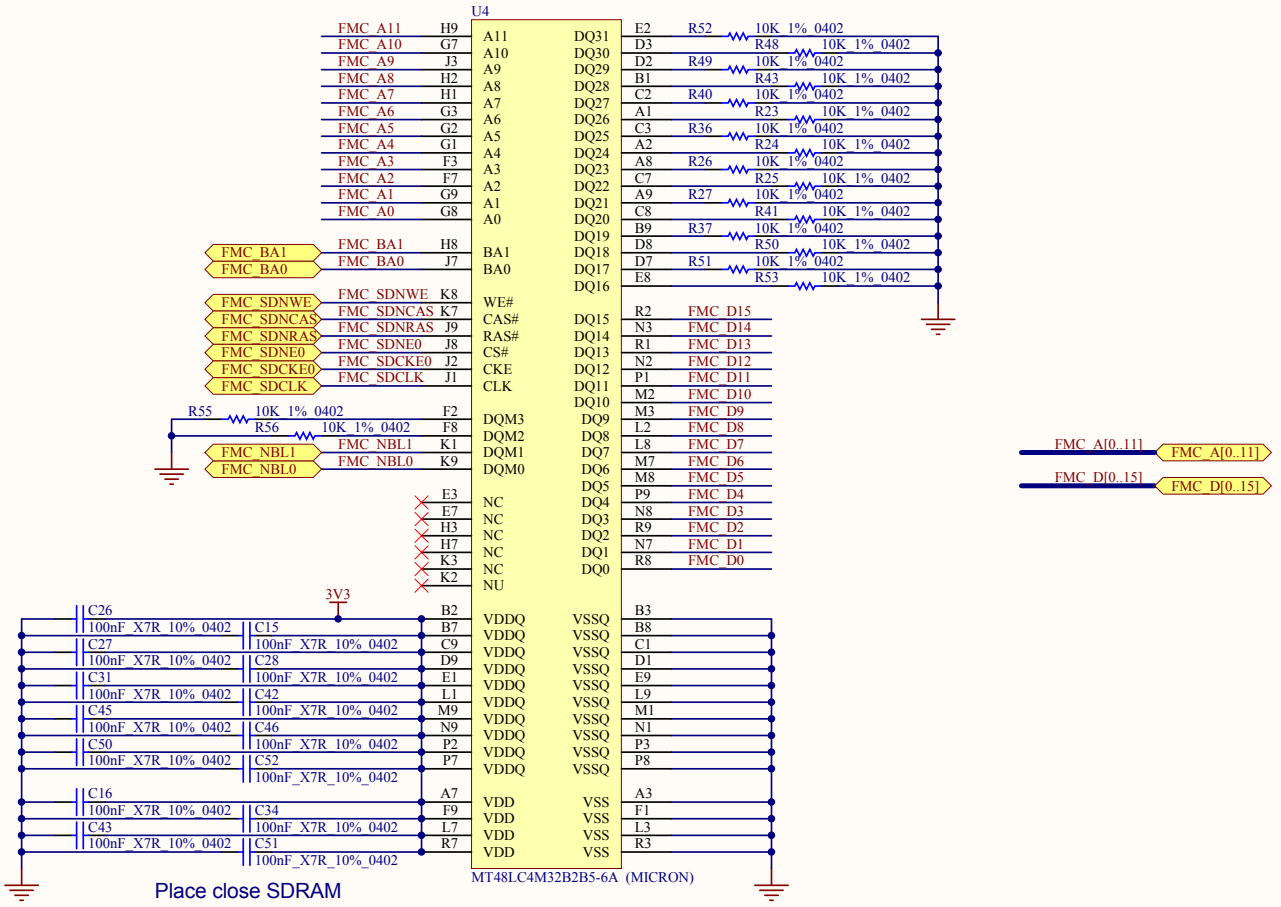
Title: **Quad SPI Flash Memory (MICRON)**			
Project: **STM32F746G-DISCO**			
Size: A4	Reference: MB1191	Revision: B-02	
Date: 5/28/2015		Sheet: 5 of 13	

life.augmented



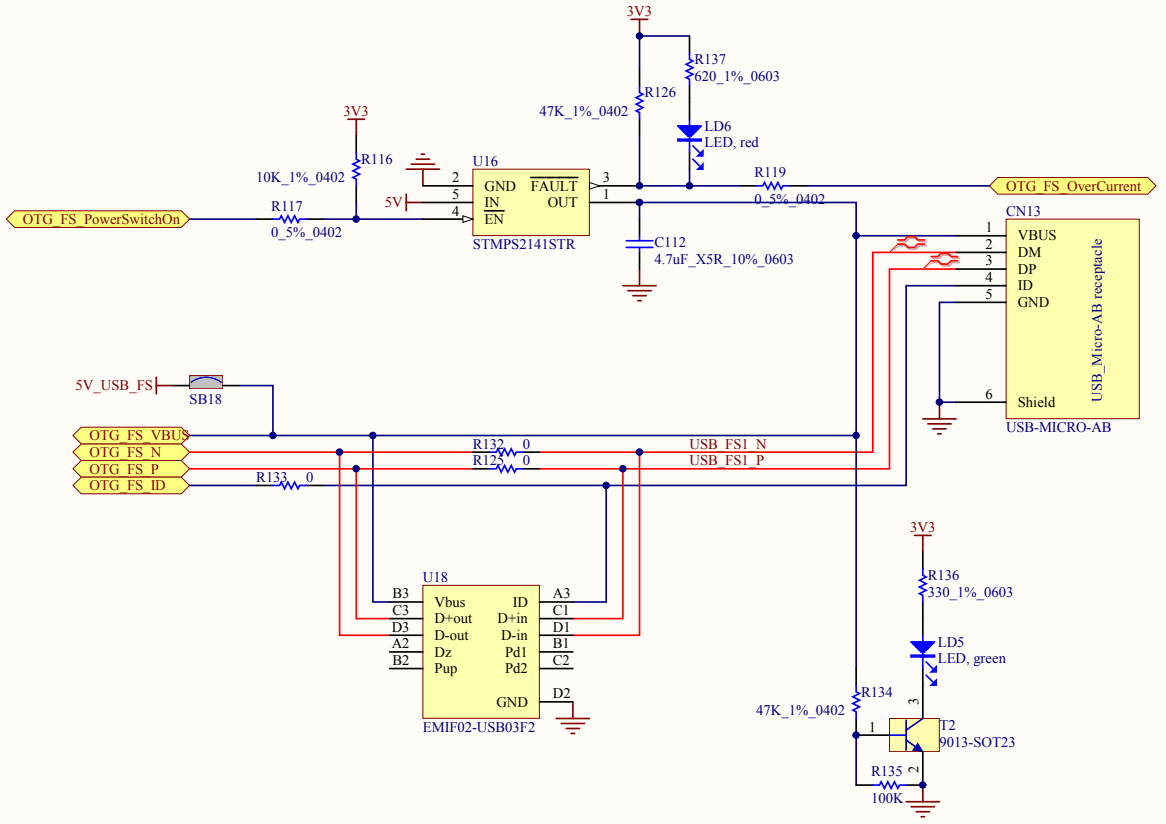
Title: **Arduino UNO connectors**		
Project: **STM32F746G-DISCO**		
Size: A4	Reference: MB1191	Revision: B-02
Date: 5/28/2015		Sheet: 6 of 13

life.augmented



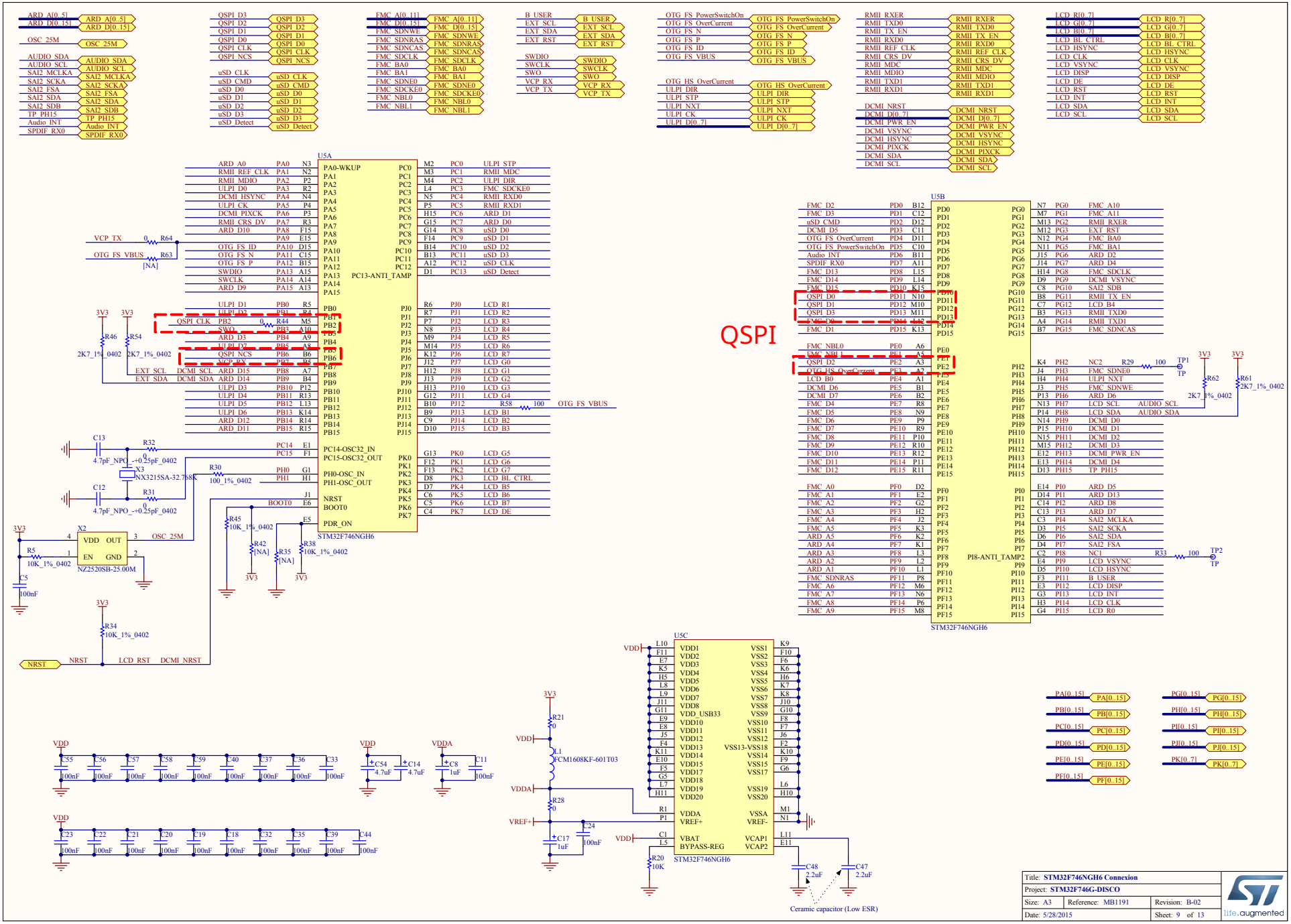
Title: **SDRAM (MICRON)**			
Project: **STM32F746G-DISCO**			
Size: A4	Reference: MB1191	Revision: B-02	
Date: 5/28/2015		Sheet: 7 of 13	

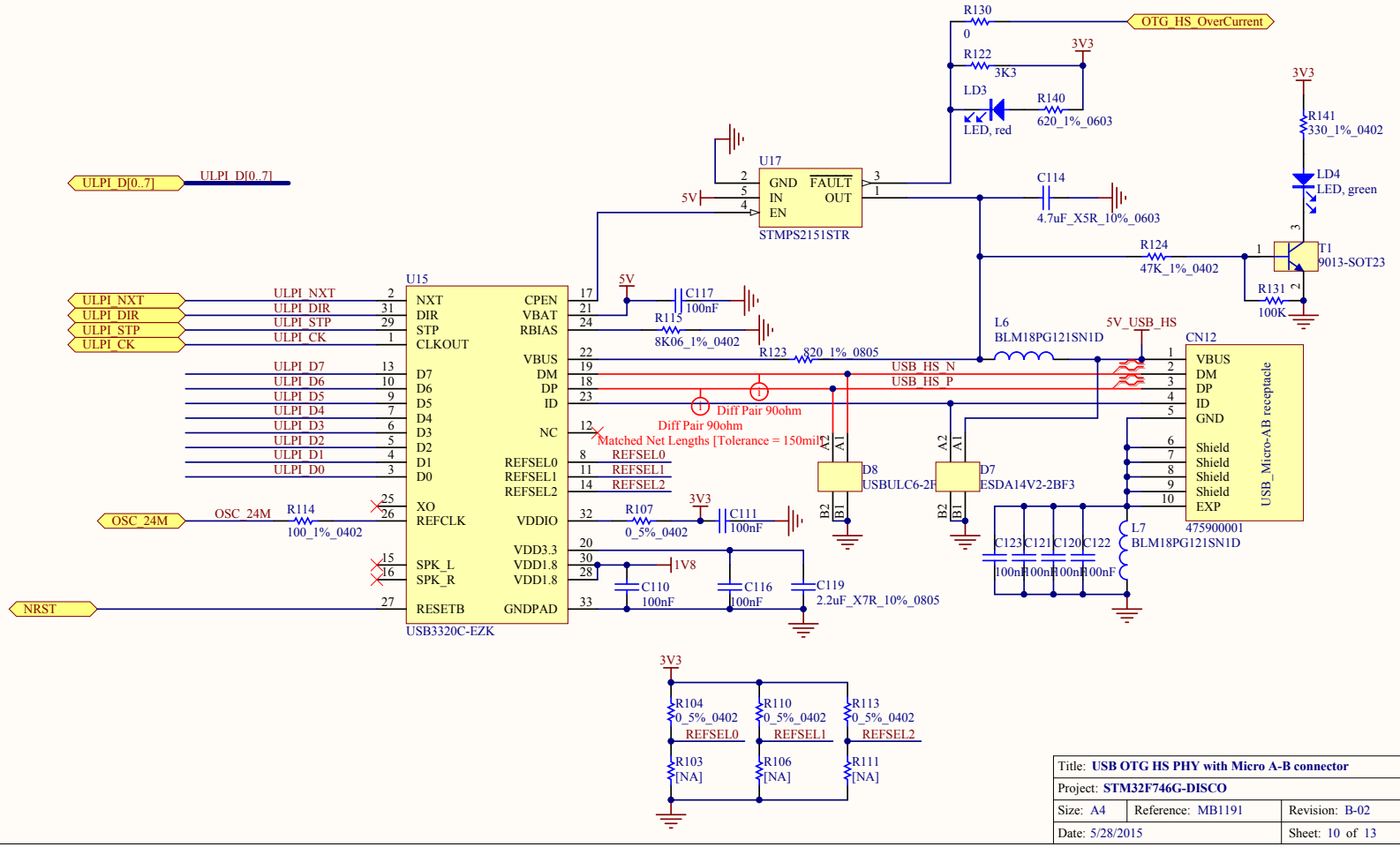
life.augmented



Title: **USB OTG FS with Micro A-B connector**			
Project: **STM32F746G-DISCO**			
Size: A4	Reference: MB1191	Revision: B-02	
Date: 5/28/2015		Sheet: 8 of 13	

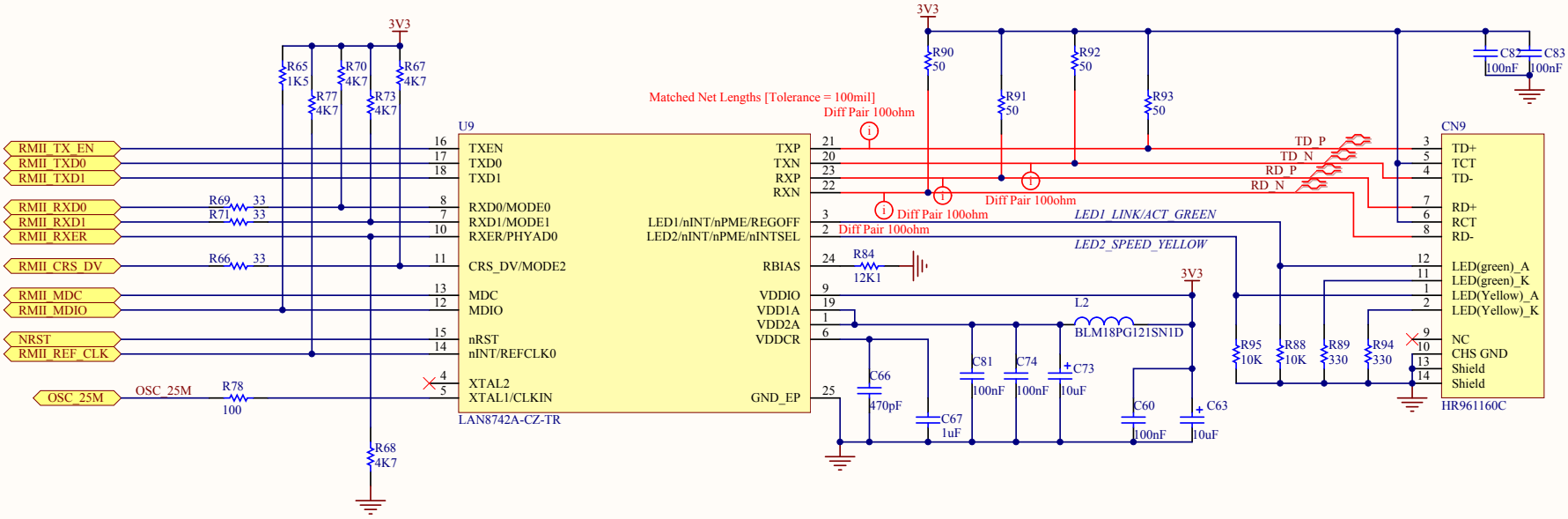
life.augmented


QSPI
U5A
U5B
U5C
STM32F746NGH6
NZ2520SB-25.00M
NX3215SA-32.768K
FCM1608KF-601T03
Ceramic capacitor (Low ESR)
Title: STM32F746NGH6 Connexion
Project: STM32F746G-DISCO
Size: A3
Reference: MB1191
Revision: B-02
Date: 5/28/2015
Sheet: 9 of 13
life.augmented



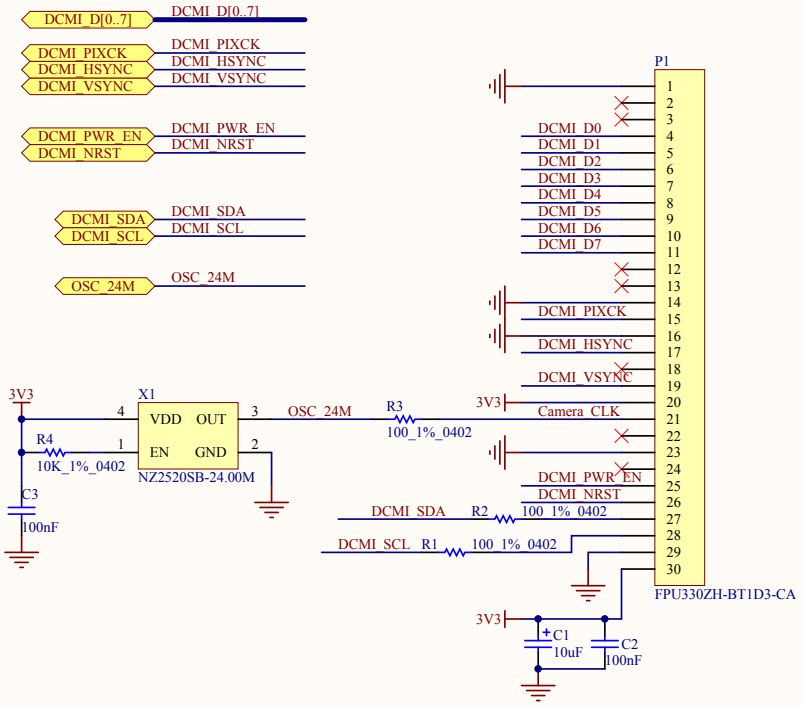
Title: **USB OTG HS PHY with Micro A-B connector**		
Project: **STM32F746G-DISCO**		
Size: A4	Reference: MB1191	Revision: B-02
Date: 5/28/2015		Sheet: 10 of 13

life.augmented



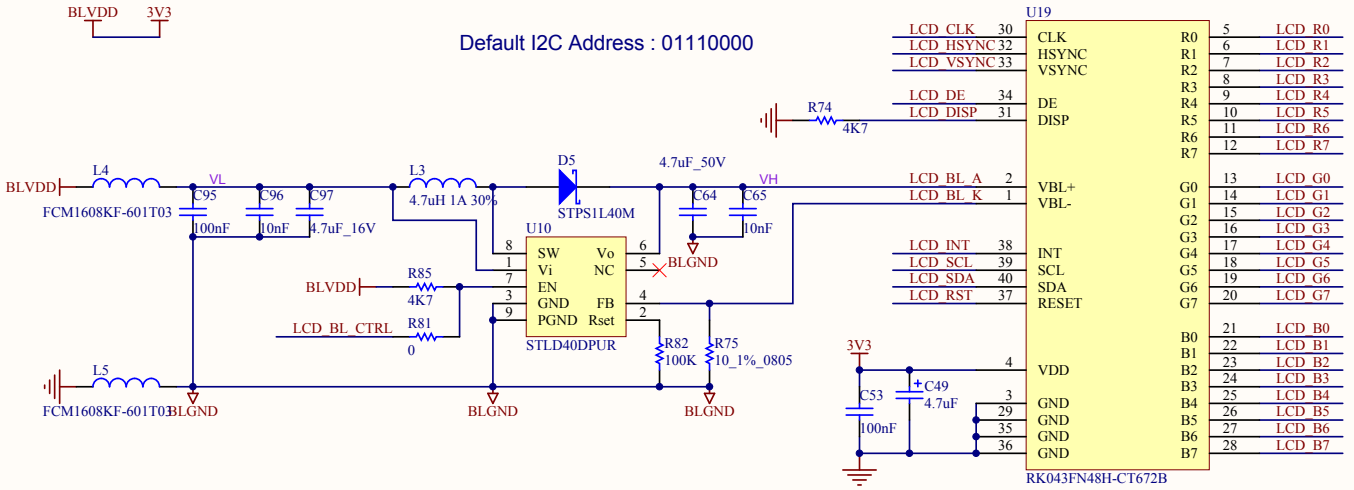
Title: **Ethernet PHY with RJ45 connector**		
Project: **STM32F746G-DISCO**		
Size: A4	Reference: MB1191	Revision: B-02
Date: 5/28/2015		Sheet: 11 of 13

life.augmented

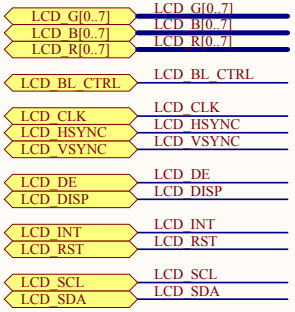


Title: **External Camera connector**			
Project: **STM32F746G-DISCO**			
Size: A4	Reference: MB1191	Revision: B-02	
Date: 5/28/2015		Sheet: 12 of 13	

life.augmented



Backlight driver & PFC of LCD panel



Title: **LCD 4.3" with Capacitive Touch**		
Project: **STM32F746G-DISCO**		
Size: A4	Reference: MB1191	Revision: B-02
Date: 5/28/2015		Sheet: 13 of 13

life.augmented